

eyes. Depending upon intended application, laser dazzlers come in a variety of package styles, mounting configurations and performance specifications. These devices can be handheld or weapon-mountable for versatility, convenience and ease of use. Some laser dazzlers have adjustable beam divergence that allows these to vary the spot size. While using a broader beam allows these a larger swath path needed to effectively produce a tactical advantage, tighter beams allow longer range and increased efficacy at longer distances.

Choice of operating parameters such as laser power, spot size at the target, laser power density and others are driven by the nature of deployment. Beam-shaping and directing optics are designed to achieve the desired value of nominal ocular hazard distance (NOHD) and a laser power density that does not exceed the maximum permissible exposure (MPE) figure dictated by American National Standards Institute (ANSI) standards for eye safety.

MPE, expressed as power density, is dependent on wavelength and exposure time. At 532nm, maximum permissible power density equals 2.5mW/cm² for 0.25-second exposure and 1.0mW/cm² for an exposure time of ten seconds.

Blue-green region is the chosen wavelength band as the human eye is most sensitive to green light. Relative spectral response (Fig. 1) is used in determining lumens of a light source and thus how bright the light will appear in terms of lux. For a red light to appear as bright as a green light, the red light must emit more radiation than the green light because humans are not as sensitive to red light.

Most-commonly-employed wavelength for the purpose is 532nm, usually generated by using either laser diodes or frequency

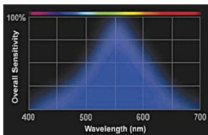


Fig. 1: Response of human eye to different wavelengths



Fig. 2: Laser dazzler for close-combat operation



Fig. 3: Vehicle-mounted laser dazzler for crowd-control application



Fig. 4: Laser dazzler in homeland security



Fig. 5: Protection of ships from asymmetric threats

doubling Nd-YAG lasers. These devices usually produce a randomly-pulsed output in the range of 10Hz to 20Hz, riding a DC level for better overall effect. DC level is usually kept at 30 per cent to 50 per cent of peak intensity level. Night-time maximum operational range is typically three to four times the maximum day-time range.

Potential applications

Non-lethal weapons act as a force multiplier, enabling friendly forces to discourage, delay or prevent hostile action. A laser dazzler is a non-lethal weapon specifically designed for applications where subject vision impairment is to be achieved at a distance ranging from few tens of metres to several kilometres in bright ambient conditions. These are particularly effective in situations where use of lethal force is not preferred; examples being limiting escalation and temporarily disabling facilities and equipment.

Laser based less-lethal weapons such as laser dazzlers can be used for counter-insurgency, anti-terrorism, counter-sniper, self-defence (Fig. 2) and crowd-control (Fig. 3) and infrastructure-protection applications.

Laser dazzlers are emerging internationally as the new non-lethally alternative to lethal force for law enforcement, homeland security (Fig. 4), border patrol, coastal protection, ship defence from asymmetric threats (Fig. 5), aircraft defence against shoulder-fired missiles (Fig. 6), infrastructure protection and host of other low-intensity conflict scenarios.

Such vehicles are usually equipped with remotely-controlled weapon stations for a wide range of applications. Technology of the short-range laser dazzler and crowd-control laser dazzler is more or less similar, except that the latter system employs a larger